

THE EFFECT OF ADDITIVES ON THE IRRADIATION BEHAVIOUR OF UO_2

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Received 14 March 1975

Samples of UO_2 doped with small amounts of Nb_2O_5 or La_2O_3 , and having various grain sizes, have been irradiated at 1500°C to 0.1% FIMA. At this low burn-up, gas release and swelling measurements show no dependence on dopant, but the Booth model prediction of swelling proportional to reciprocal grain size has been verified. Gas release does not fit the simple Booth model at the low releases measured, and shows a dependence on sample density, and hence surface area only. A model has been derived to explain these results. The rare gas diffusion coefficient in UO_2 at 1500°C has been measured to be $1.6 \times 10^{-19} \text{ m}^2/\text{s}$.

Des échantillons d' UO_2 dopés par de petites quantités de Nb_2O_5 ou de La_2O_3 et possédant des tailles de grains différentes ont été irradiés à 1500°C à 0,1% FIMA. Pour ce faible taux de combustion le dégagement des gaz et les mesures de gonflement ne montrent pas d'effet du dopant, mais la prédiction par le modèle de Booth d'un gonflement proportionnel à l'inverse de la taille du grain a été vérifiée. Le dégagement de gaz ne satisfait pas entièrement au modèle de Booth pour les faibles dégagements de gaz mesurés mais dépend de la densité des échantillons et par suite dépend seulement de l'aire superficielle. Un modèle en a été déduit afin d'expliquer ces résultats. Le coefficient de diffusion des gaz rares dans UO_2 à 1500°C a été mesuré comme étant égal à $1,6 \times 10^{-19} \text{ m}^2/\text{sec}$.

UO_2 verschiedener Korngrösse wurde mit geringen Mengen Nb_2O_5 oder La_2O_3 dotiert und bei 1500°C bis zu 0,1% FIMA bestrahlt. Bei diesem niedrigen Abbrand ist eine Abhängigkeit der Gasfreisetzung und des Schwellens vom Dotierungsmittel nicht erkennbar, die Voraussage eines der reziproken Korngrösse proportionalen Schwellens nach den Boothschen Modell wird bestätigt. Die Gasfreisetzung bei den niedrigen gemessenen Freisetzungswerten wird durch das Boothschen Modell nicht belegt, sie ist nur abhängig von der Dichte und somit von der spezifischen Oberfläche der Probe. Ein Modell zur Erklärung dieser Ergebnisse wurde abgeleitet. Der Diffusionskoeffizient für die Edelgase im UO_2 wurde bei 1500°C gemessen, er beträgt $1,6 \times 10^{-19} \text{ m}^2/\text{s}$.

1. Introduction

Booth [1] has derived an expression relating the diffusion coefficient, D , to the fractional release of stable fission gas from nuclear fuel during irradiation, F , which is

$$F = 1 - \frac{6a^2}{\pi^4 Dt} \sum_{n=1}^{\infty} \frac{1}{n^4} \left\{ 1 - \exp\left(\frac{-n^2 \pi Dt}{a^2}\right) \right\}$$

where t is the time and a is the radius of a hypothetical, spherical diffusion volume.

Speight [2] has modified the Booth equation to allow for the effects of irradiation-induced re-resolution of gas atoms from intra-granular bubbles, by introducing a modified diffusion coefficient, D' , which is given by

$$D' = D \frac{b}{(b + g)}$$

where b and g are re-resolution constants. This modification does not alter the form of the diffusion equations, and the modified diffusion coefficient has been used throughout this paper.

The Booth model is applied to gas release and swelling by considering the diffusion sphere to be equivalent to a grain in the fuel. This approximation is valid for swelling when this is due to intergranular gas bubbles only and no interlinked porosity exists, and applies to gas release only after interlinked porosity has formed. Turnbull and Tucker [3] have derived expressions for gas swelling after interlinked porosity has occurred, and a simple model for low quantity gas release has been

developed in this paper which gives the fractional release, F , as

$$F = \frac{S}{V} \cdot \frac{4}{3} \sqrt{\frac{Dt}{\pi}}$$

where S/V is the surface-to-volume ratio of the sample.

The Booth model predicts $F \propto 1/a$ for small gas release, and this leads to the prediction that the rare-gas swelling rate of nuclear fuels is proportional to the inverse grain size of the fuel.

These considerations suggest that a reduction in the rate of swelling and gas release would be achieved if either the diffusion coefficient were to be reduced or a dense fuel with large grain size were used. Miekeley and Felix [4] show that the gas diffusion coefficient is lowered in hypostoichiometric UO_{2-y} and is increased in hyperstoichiometric UO_{2+x} . To try to utilise this effect, samples of UO_2 were prepared which had been doped with 0.1 at.% and 1.0 at.% of La_2O_3 or Nb_2O_5 to simulate the hypo and hyper-stoichiometric conditions respectively. One further sample of 1.0 at.% Nb_2O_5 -doped material was annealed at 1550°C for 16 hours in argon to produce a large-grained sample (grain size approx. $80\text{ }\mu\text{m}$). These five samples, together with one reference sample of undoped UO_2 , were irradiated to 0.1% FIMA at 1500°C to observe the effects of doping and grain size on the gas release and swelling properties of uranium dioxide.

2. Experimental details

Six small, cylindrical samples 10mm long by 2.3mm diameter were prepared from unenriched ura-

nium dioxide doped with small quantities of aliovalent rare earth oxides, which had been chosen for their low thermal neutron cross-sections. The composition, density and original grain size of these samples are shown in table 1.

All the samples were sealed into double-encased, molybdenum alloy (TZM) capsules under a helium atmosphere at approximately half atmospheric pressure. The capsules were irradiated in a high temperature rig in the UKAEA reactor DIDO under a flowing atmosphere of $\text{He} + 2\% \text{H}_2$, which would maintain the undoped UO_2 sample at stoichiometry by diffusion of hydrogen into the specimen enclosure through the capsule walls. The hydrogen in the atmosphere would also maintain the O/M ratio below 2 for the La_2O_3 -doped samples [5], but probably reduced the Nb_2O_5 -doped samples to give a stoichiometric O/M ratio of 2 or possibly slightly below [6]. The irradiation temperature was nominally 1500°C at the sample surface and approximately 50°C higher at the centre of the specimen due to fission heating. All the samples were irradiated to approximately 0.1% FIMA in a thermal neutron flux of $2.2 - 2.6 \times 10^{17} \text{m}^{-2}/\text{s}$. The full irradiation details are given in table 2.

Before irradiation, the dimensions of each specimen were accurately measured using a dial gauge, and its density determined by an immersion technique [7]. The pre-irradiation microstructure was also determined.

When the irradiation was completed, the quantities of ^{133}Xe and ^{85}Kr released into the capsules from each specimen were measured by gamma-counting. The dimensions and densities were remeasured before the samples were cut into 1mm thick slices. One slice from each sample was polished for optical examination,

Table 1
Composition and physical characteristics of the doped UO_2 samples

Sample number	Dopant (mol%)	Density (Mg/m^3)	Density (% theoretical)	Grain size (μm)
1	0	10.66	97.3	7
2	0.1% Nb_2O_5	10.59	96.7	7
3	1.0% Nb_2O_5	10.16	93.2	28
4a)	1.0% Nb_2O_5	10.43	95.7	80
5	0.1% La_2O_3	10.79	98.5	16
6	1.0% La_2O_3	10.48	96.1	11

a) Sample 4 was annealed in argon for 16 hours at 1550°C before irradiation.

Table 2
Irradiation temperatures in °C

Sample number	Reactor cycle (1 cycle $\sim 2 \times 10^6$ s)					Neutron flux (10^{17} n/m ² s)
	1	2	3	4	5	
1	1492	1537	1463	1453	1466	2.28
2	1483	1498	1478	1468	1463	2.54
3	1513	1528	1508	1498	1493	2.61
4	1513	1528	1508	1498	1493	2.58
5	1462	1507	1433	1423	1436	2.19
6	1492	1537	1463	1453	1466	2.24

while further pieces were used for scanning electron microscopy and for gas retention analysis.

3. Results

3.1. Swelling

After irradiation the samples were visually examined showing the surface finish apparently unchanged, which indicated that no gross distortion or chemical interaction with the cladding had occurred. Swelling measurements were made by two separate methods, firstly by measuring the external dimensional changes, with a correction made for material that had evaporated from the sample during irradiation, and secondly by an immersion density technique.

Changes in both the length and diameter of the specimens were measured, but owing to some ovality, the diametral measurements have a spread which completely masked the small changes due to swelling. Length measurements were only possible on four of the samples, both of the other two were broken during irradiation into two pieces. The volume swelling calculated for these four specimens correspond broadly to the density determination, but all are higher, which could indicate some surface porosity formation during irradiation. The density swelling results are plotted in fig. 1 as a function of reciprocal grain size. A correction to the measured values of samples 1, 2, 5 and 6 has been made to try to allow for variations in temperature during irradiation and bring them all to a calculated temperature of $\approx 1500^\circ\text{C}$. The correction was made by using the Booth model, which gives the number of gas atoms, $\bar{N}$, on a grain face as proportional to

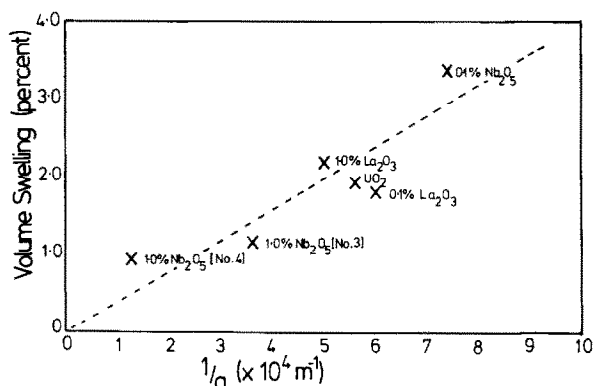


Fig. 1. Plot of swelling against the reciprocal grain size. Swelling measurements were made by an immersion density technique and have been corrected to allow for variation in irradiation temperature.

the square root of the diffusion coefficient and also the relation

$$n \propto V^{2/3}$$

where V is the volume of a grain-boundary gas bubble containing n atoms. The activation energy for D is assumed to be 90 kcal per mole and the number of gas bubbles on the grain boundary is conservatively assumed to be constant. With this correction, all size results lie on a straight line passing through the origin of fig. 1, which is as expected from the Booth model if it is assumed that the rare gas diffusion coefficient is unchanged by the aliovalent additives. The uncorrected swelling results are also tabulated in table 3.

3.2. Gas release

The gas release results are plotted in figs. 2 and 3

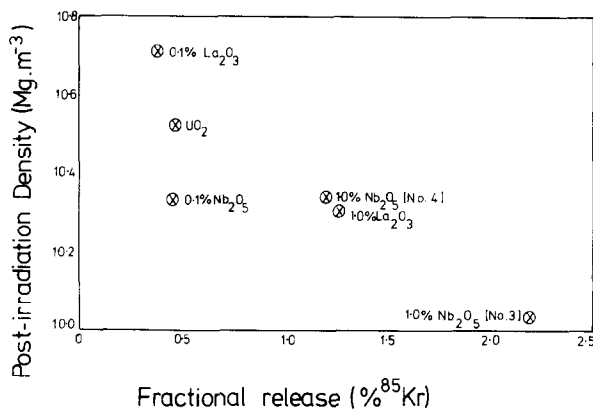


Fig. 2. Plot of ^{85}Kr release against density.

as a function of density. The amounts released are small, indicating that no interlinked porosity exists, and no correlation between these results and grain size could be found. Use of the Booth equivalent sphere model leads to anomalous results in this experiment, for the diffusional sphere has a calculated volume greater than the actual volume of the sample. To remove this anomaly an expression has been derived in the Appendix which gives the fractional release of stable rare gasses, F , as a function of the surface-to-volume ratio of the sample, S/V .

$$F = \frac{4}{3} \frac{S}{V} \sqrt{(Dt/\pi)}.$$

Surface area measurements were not made on the irradiated specimens and BET measurements [8] on

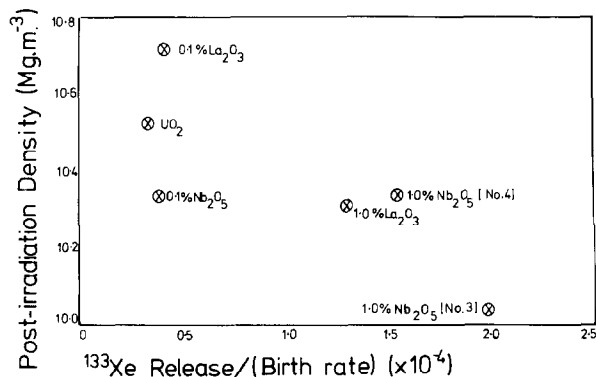


Fig. 3. Plot of ^{133}Xe release against density. Both this and fig. 2 demonstrate that the observed gas releases are independent of the dopant used, but that low release is associated with high density, and hence surface area, only.

archive material were not sufficiently sensitive to measure the very small areas involved. The expression for F does lead to the prediction that less dense material, which would be porous and hence have a larger surface area, will release a larger fraction of the rare gas generated within the specimen, as is shown in figs. 2 and 3.

An upper limit to the value of D can be calculated from the results, however, if the release from the most dense sample is used and the surface area is assumed to be equal to the geometrical surface area; then a value of $D \approx 7 \times 10^{-19} \text{ m}^2 \text{ s}$ is derived. A surface roughness factor of 2 would reduce this value of D to $1.7 \times 10^{-19} \text{ m}^2 \text{ s}$ which is very close to the value found by Cornell [9].

The release of short-lived rare gas isotopes is given in a similar manner by

$$\frac{R}{B} = \frac{S}{V} \sqrt{D} \lambda^{-n}$$

where $n = \frac{1}{2}$, R/B is the release to birth rate ratio and λ is the half life of the rare gas isotope. Substituting the value of D found for stable gas release, does not give consistency, but if n is assumed to be less than $\frac{1}{2}$, ($n \approx 0.26$) this inconsistency is removed. The result that rare gas release is not given by a $\lambda^{-\frac{1}{2}}$ dependence, but a λ^{-n} ($n < \frac{1}{2}$) dependence has often been observed, but no complete explanation has yet been found [10,11].

3.3. Microstructure

Post-irradiation examination of the specimens showed that only the three samples with the smallest initial grain size had undergone measurable grain growth during irradiation. It is known that short exposures to irradiation are sufficient to greatly reduce the rate of grain growth in UO_2 [7, 12]. If it is assumed that all the grain growth measured in this experiment occurred up to a time t , after which the burn-up was sufficient to prevent grain growth, then the standard thermal activation equation for grain growth can be used [13]:

$$d_1^2 - d_0^2 = \frac{Kt^n \exp(-Q/RT)}{T} \quad (1)$$

where d_1 and d_0 are the final and initial grain sizes, t is the sintering time, T the sintering temperature, Q

Table 3
Uncorrected swelling measurements

Sample number	Volume swelling (%)	%wt loss	Density swelling (%)	Final grain size (μm)
1	1.48	0.38	1.36	18.0
2	2.60	0.33	2.39	13.5
3	—	1.50	1.21	28.0
4	1.10	0.80	0.91	80.0
5	0.79	0.40	0.83	16.5
6	—	0.60	1.61	20.0

the activation energy for sintering and K , n and R are constants. In this experiment, time and temperature are also sensibly constant and so equation (1) can be simplified to give:

$$d_1^2 - d_0^2 = B \exp(-Q) \quad (2)$$

where B is a constant.

Evaluation of the left hand side of equation (2) from measurements made on the three specimens that showed significant grain growth (UO_2 , $\text{UO}_2/0.1$ at.% Nb_2O_5 , $\text{UO}_2/1.0$ at.% La_2O_3) gave values that could be considered to be constant within the experimental error limits. This constant is such that no measurable grain growth could be expected for the two large grain-size samples 3 and 4, whilst the fact that no grain growth was observed for sample 5 ($\text{UO}_2/0.1$ at.% La_2O_3) can be explained by consideration of the lower than average temperature experienced by this sample during irradiation (table 2).

The results indicate that Q is constant for all the six samples under the present experimental conditions of temperature and atmosphere, and hence is unaffected by doping and the predicted extrinsic diffusion behaviour that aliovalent additives should produce. This behaviour is almost certainly caused by the reducing atmosphere of this experiment, though irradiation effects cannot be entirely ruled out.

Grain growth out-of-pile in a neutral atmosphere produced very different results, with 1.0 at.% Nb_2O_5 -doped material showing grain growth from $30 \mu\text{m}$ to $100 \mu\text{m}$ during an anneal at 1550°C for 16 hours in argon, while undoped UO_2 showed grain growth from $11 \mu\text{m}$ to $13 \mu\text{m}$ under the same conditions. These results show that very large-grained material can be easily produced from Nb_2O_5 -doped material, and that this grain structure is stable during irradiation, at

least under reducing conditions.

Examination of polished sections showed lenticular gas bubbles on grain boundaries and also triangular cross section tunnels on grain edges. There was no evidence of interlinked porosity and the bubble densities were low, especially for the samples with little measured swelling and the swelling expected from the observed bubble concentrations correlate well with the measured density changes.

3.4. Scanning electron microscopy

Fractured surfaces of all the specimens were examined by scanning electron microscopy; figs. 4 and 5 show micrographs of samples 1 and 2. Lenticular bubbles were seen on all the grain faces examined, and significant interlinkage was observed only on one face on sample 4 ($\text{UO}_2/1.0$ at.% Nb_2O_5). All the samples, except number 5 ($\text{UO}_2/0.1$ at.% La_2O_3), showed a proportion of the escaped gas to be residing in grain-edge bubbles, which were often seen to be on the point of interlinking to form tunnels. In common with Turnbull [7], a maximum density of gas on the grain faces was found $\sim 2 \times 10^{18}$ atoms/ m^2 , mostly residing in bubbles $\approx 1 \mu\text{m}$ diameter on the grain face. This number also gives a lower limit to the diffusion coefficient, calculated from the Booth model which gives the number of gas atoms, N , reaching unit area of a surface:

$$N = \frac{4}{3} \beta t \sqrt{\frac{Dt}{\pi}} - \frac{\beta Dt^2}{2d}$$

where β is the gas generation rate which is 4×10^{17} atoms/ m^3 s.

i.e. $D \geq 4 \times 10^{-20} \text{m}^2/\text{s}$.

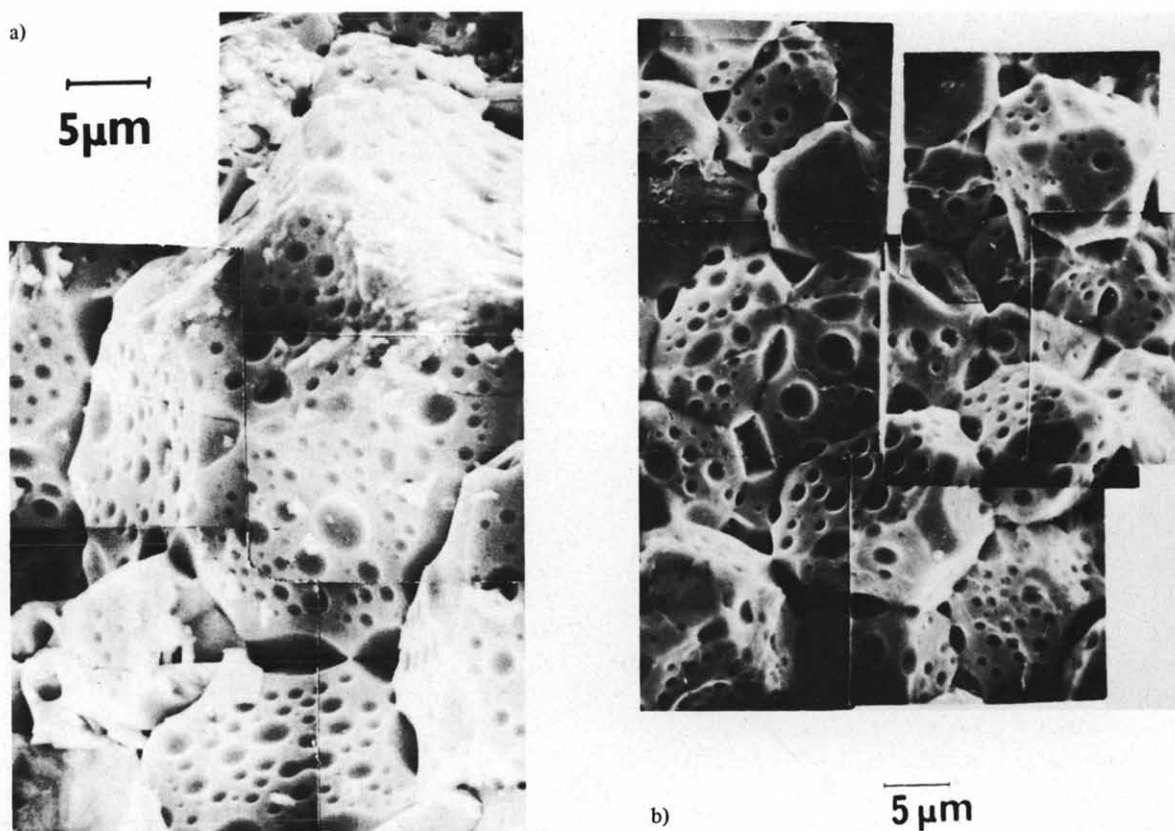


Fig. 4. Scanning electron micrographs of the fractured surfaces of (a) undoped UO_2 and (b) $0.1\text{ at } \% \text{ Nb}_2\text{O}_5$ doped UO_2 showing the similarity between the intergranular gas bubble distributions for both doped and undoped UO_2 .

Re-solution of gas from grain boundaries modifies the boundary conditions used to solve the diffusion equations, and exact solutions are not possible. However, using the methods developed by Speight [2] it can be shown that re-solution probably only alters the value of D derived above by approximately 25%.

4. Conclusions

The experimental results demonstrate that adding small quantities of aliovalent dopants to UO_2 does not directly affect the irradiation behaviour under the present conditions. No effect that could be ascribed to doping was observed on swelling, gas release or grain growth during irradiation. Indirectly, doping has had an effect in that it determined the original grain size and density of uranium dioxide and these properties in turn determine the rates of both swelling and gas

release. Further points that arose from this experiment were:

1. The rare gas diffusion coefficient has been measured at 1500°C to lie in the range 7×10^{-19} to $4 \times 10^{-20} \text{ m}^2/\text{s}$, with a probable value of $1.6 \times 10^{-19} \text{ m}^2/\text{s}$, which is the value found by Cornell [9].
2. The rate of grain growth during irradiation under a reducing atmosphere is unaffected by additives and is considerably slower than that found out-of-pile. This means that very large grains that are stable under irradiation can easily be formed by pre-irradiation treatment of UO_2 , in particular by the addition of small amounts of Nb_2O_5 before sintering.
3. The prediction of the Booth model, that swelling should be proportional to the reciprocal grain size, has been verified up to approximately 4% swelling.
4. Gas release at the low levels measured does not fit the simple Booth model and seems to be a function of density only, and hence most probably only of the

surface area of a sample. A model has been developed to give this dependence but the surface areas of the present samples are too small to allow complete verification of the model.

5. Unstable gas release does not fit the predicted $\lambda^{-\frac{1}{2}}$ dependence and a $\lambda^{-\frac{1}{4}}$ dependence has been found. Similar results have often been observed [10], but no explanation of this can be offered.

Acknowledgement

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Appendix

The Booth model is not applicable to the gas release results in this experiment, as the Booth equivalent sphere has a calculated radius such that its volume is greater than that of the sample from which the release is occurring, leading to problems in interpreting the fractional release. The problem arises from the non-ideal geometry of the present samples.

To resolve this, gas release is assumed to occur from only a small depth within the specimen, and hence a semi-infinite plane geometry is used to solve the diffusion equation

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} + \beta$$

where C is the concentration of gas atoms, β is the gas generation rate, D the diffusion coefficient, t the time and x is the distance inside the solid whose surface is at $x = 0$.

The boundary conditions are

$$C = 0 \text{ at } t = 0,$$

$$C = 0 \text{ at } x = 0 \text{ for all } t.$$

The solution is given by Carslaw and Jaeger [14] where:

$$C(x) = \left\{ \beta t + 2\beta t \left(\frac{x}{2\sqrt{Dt}} \right)^2 \right\} \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right)$$

$$+ \beta \frac{x}{2\sqrt{Dt}} \cdot \frac{2t}{\sqrt{\pi}} \cdot \exp \left(\frac{-x}{2\sqrt{Dt}} \right)^2 - 2\beta t \left(\frac{x}{2\sqrt{Dt}} \right)^2.$$

The number of gas atoms, N , that have escaped from unit surface area at time t is given by

$$N = \int_0^\infty (C_\infty - C(x)) dx,$$

where C_∞ is the concentration at infinity and is given by:

$$C_\infty = \beta t.$$

Substituting $y = x/2\sqrt{Dt}$.

$$N = 2\sqrt{Dt} \beta t \int_0^\infty \left\{ \operatorname{erfc}(y) + 2y^2 \operatorname{erfc}(y) - \frac{2}{\sqrt{\pi}} y e^{-y^2} \right\} dy$$

$$\begin{aligned} &= 2\beta t \sqrt{Dt} \left\{ \operatorname{ierfc}(0) + \left[\frac{2}{3} y^2 \operatorname{erfc}(y) \right]_0^\infty \right. \\ &\quad \left. - \int_0^\infty \left(-\frac{2y^3}{3} \frac{2}{\sqrt{\pi}} e^{-y^2} \right) dy + \left[\frac{2}{\sqrt{\pi}} \frac{e^{-y^2}}{2} \right]_0^\infty \right\} \\ &= 2\beta t \sqrt{Dt} \left\{ \frac{1}{\sqrt{\pi}} + \frac{\Gamma(2)}{2} \cdot \frac{4}{3\sqrt{\pi}} - \frac{1}{\sqrt{\pi}} \right\} \end{aligned}$$

$$N = \frac{4}{3} \beta t \frac{\sqrt{Dt}}{\sqrt{\pi}}.$$

The fraction of gas released, F , from a sample of surface area S and volume V is given by

$$F = \frac{S}{\beta t V} \cdot N = \frac{4}{3} \frac{S}{V} \sqrt{\frac{Dt}{\pi}}. \quad (1)$$

For the case of a sphere radius a , this equation reduces to give the Booth release approximation for small releases:

$$F_{\text{Booth}} = 4 \frac{\sqrt{Dt}}{\sqrt{\pi} a^2}.$$

Equation (1) is only applicable to geometries other than the semi-infinite plane when diffusion occurs from a distance y inside the solid where y is very much less than the dimensions of the specimen. This distance is given by:

$$y = 1.5 \sqrt{Dt}$$

when 99% of the measured release has occurred from inside this distance. In the present experiment $y \approx 2\mu\text{m}$.

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